

```
import os
import zipfile
from tensorflow.keras.applications import VGG16
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Flatten, Dense
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.preprocessing.image import ImageDataGenerator
```

```
!kaggle datasets download -d paultimothymooney/chest-xray-pneumonia
```

```
Dataset URL: https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia
License(s): other
Downloading chest-xray-pneumonia.zip to /content
100% 2.29G/2.29G [00:11<00:00, 216MB/s]
```

```
with zipfile.ZipFile("chest-xray-pneumonia.zip", "r") as zip_ref:
    zip_ref.extractall("chest_xray_dataset")
```

```
train_dir = "chest_xray_dataset/chest_xray/train"
val_dir = "chest_xray_dataset/chest_xray/val"
test_dir = "chest_xray_dataset/chest_xray/test"
```

```
train_datagen = ImageDataGenerator(rescale = 1.0/255,
                                   rotation_range = 20,
                                   width_shift_range = 0.2,
                                   height_shift_range=0.2,
                                   shear_range=0.2, zoom_range = 0.2,
                                   horizontal_flip = True)
```

```
val_datagen = ImageDataGenerator(rescale=1.0/255)
```

```
train_generator = train_datagen.flow_from_directory(train_dir, target_size=(224, 224), batch_size = 32, class_mode = 'binary')
val_generator = val_datagen.flow_from_directory(val_dir, target_size = (224, 224), batch_size = 32, class_mode='binary')
```

```
Found 5216 images belonging to 2 classes.
Found 16 images belonging to 2 classes.
```

```
base_model = VGG16(weights = "imagenet", include_top = False, input_shape = (224, 224, 3))
for layer in base_model.layers:
    layer.trainable = False

x = Flatten()(base_model.output)
x = Dense(256, activation = 'relu')(x)
output = Dense(1, activation = 'sigmoid')(x)

model = Model(inputs = base_model.input, outputs = output)
model.compile(optimizer = Adam(learning_rate = 0.0001), loss = 'binary_crossentropy', metrics = ['accuracy'])
model.summary()
```

Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/vgg16/vgg16_weights_tf_dim_ordering_tf_kernels/58889256/58889256_0s_0us/step

Model: "functional"

Layer (type)	Output Shape	Param #
input_layer (InputLayer)	(None, 224, 224, 3)	0
block1_conv1 (Conv2D)	(None, 224, 224, 64)	1,792
block1_conv2 (Conv2D)	(None, 224, 224, 64)	36,928
block1_pool (MaxPooling2D)	(None, 112, 112, 64)	0
block2_conv1 (Conv2D)	(None, 112, 112, 128)	73,856
block2_conv2 (Conv2D)	(None, 112, 112, 128)	147,584
block2_pool (MaxPooling2D)	(None, 56, 56, 128)	0
block3_conv1 (Conv2D)	(None, 56, 56, 256)	295,168
block3_conv2 (Conv2D)	(None, 56, 56, 256)	590,080
block3_conv3 (Conv2D)	(None, 56, 56, 256)	590,080
block3_pool (MaxPooling2D)	(None, 28, 28, 256)	0
block4_conv1 (Conv2D)	(None, 28, 28, 512)	1,180,160
block4_conv2 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_conv3 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_pool (MaxPooling2D)	(None, 14, 14, 512)	0
block5_conv1 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv2 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv3 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_pool (MaxPooling2D)	(None, 7, 7, 512)	0
flatten (Flatten)	(None, 25088)	0
dense (Dense)	(None, 256)	6,422,784
dense_1 (Dense)	(None, 1)	257

Total params: 21,137,729 (80.63 MB)

Trainable params: 6,423,041 (24.50 MB)

Non-trainable params: 14,714,688 (56.13 MB)

```
model.fit(train_generator, validation_data=val_generator, epochs=10, steps_per_epoch=len(train_generator), validation_steps=len(val_generator),
test_datagen = ImageDataGenerator(rescale=1.0 / 255)
test_generator = test_datagen.flow_from_directory(test_dir, target_size = (224, 224), batch_size=32, class_mode = 'binary')
```

```
Epoch 1/10
163/163 ————— 139s 752ms/step - accuracy: 0.8869 - loss: 0.2650 - val_accuracy: 0.8125 - val_loss: 0.3898
Epoch 2/10
163/163 ————— 113s 692ms/step - accuracy: 0.9385 - loss: 0.1593 - val_accuracy: 0.6875 - val_loss: 0.5985
Epoch 3/10
163/163 ————— 112s 685ms/step - accuracy: 0.9379 - loss: 0.1494 - val_accuracy: 0.8125 - val_loss: 0.4686
Epoch 4/10
163/163 ————— 113s 693ms/step - accuracy: 0.9452 - loss: 0.1352 - val_accuracy: 0.8125 - val_loss: 0.2915
Epoch 5/10
163/163 ————— 111s 684ms/step - accuracy: 0.9484 - loss: 0.1341 - val_accuracy: 0.6875 - val_loss: 0.7488
Epoch 6/10
163/163 ————— 112s 686ms/step - accuracy: 0.9482 - loss: 0.1348 - val_accuracy: 0.6875 - val_loss: 0.4592
Epoch 7/10
163/163 ————— 112s 688ms/step - accuracy: 0.9565 - loss: 0.1099 - val_accuracy: 0.6875 - val_loss: 0.6640
Epoch 8/10
163/163 ————— 144s 695ms/step - accuracy: 0.9571 - loss: 0.1076 - val_accuracy: 0.6875 - val_loss: 0.8662
Epoch 9/10
163/163 ————— 112s 684ms/step - accuracy: 0.9521 - loss: 0.1275 - val_accuracy: 0.6875 - val_loss: 1.0462
Epoch 10/10
163/163 ————— 140s 670ms/step - accuracy: 0.9584 - loss: 0.1104 - val_accuracy: 0.7500 - val_loss: 0.3881
Found 624 images belonging to 2 classes.
```

```
test_loss, test_accuracy = model.evaluate(test_generator)
print(f"Test Loss: {test_loss:.4f}, Test Accuracy: {test_accuracy:.4f}")
```

```
20/20 ————— 7s 300ms/step - accuracy: 0.9103 - loss: 0.2651
Test Loss: 0.2651, Test Accuracy: 0.9103
```

Transfer Learning Strategy 2

```

train_dir = "chest_xray_dataset/chest_xray/train"
val_dir = "chest_xray_dataset/chest_xray/val"
test_dir = "chest_xray_dataset/chest_xray/test"
train_datagen = ImageDataGenerator(rescale=1.0 / 255, rotation_range=20, width_shift_range=0.2, height_shift_range=0.2, shear_range=0.2, zoom_range=0.2, horizontal_flip=True, vertical_flip=True)
val_datagen = ImageDataGenerator(rescale=1.0 / 255)

```

```

train_generator = train_datagen.flow_from_directory(train_dir, target_size=(224, 224), batch_size=32, class_mode='binary')
val_generator = val_datagen.flow_from_directory(val_dir, target_size=(224, 224), batch_size=32, class_mode='binary')

```

```
base_model = VGG16(weights='imagenet', include_top=False, input_shape=(224, 224, 3))
```

```

for layer in base_model.layers:
    layer.trainable = False

```

```

x = Flatten()(base_model.output)
x = Dense(256, activation='relu')(x)
output = Dense(1, activation='sigmoid')(x)
model = Model(inputs=base_model.input, outputs=output)
model.compile(optimizer=Adam(learning_rate=0.001), loss='binary_crossentropy', metrics=['accuracy'])
model.summary()

```

Found 5216 images belonging to 2 classes.
Found 16 images belonging to 2 classes.

Model: "functional_1"

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 224, 224, 3)	0
block1_conv1 (Conv2D)	(None, 224, 224, 64)	1,792
block1_conv2 (Conv2D)	(None, 224, 224, 64)	36,928
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block3_conv3 (Conv2D)	(None, 56, 56, 256)	590,080
block3_pool (MaxPooling2D)	(None, 28, 28, 256)	0
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block4_conv3 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_pool (MaxPooling2D)	(None, 14, 14, 512)	0
block5_conv1 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv2 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv3 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_pool (MaxPooling2D)	(None, 7, 7, 512)	0
flatten_1 (Flatten)	(None, 25088)	0
dense_2 (Dense)	(None, 256)	6,422,784
dense_3 (Dense)	(None, 1)	257

Total params: 21,137,729 (80.63 MB)

Trainable params: 6,423,041 (24.50 MB)

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```

model.fit(train_generator, validation_data=val_generator, epochs=10, steps_per_epoch=len(train_generator), validation_steps=len(val_generator))
test_datagen = ImageDataGenerator(rescale=1.0 / 255)
test_generator = test_datagen.flow_from_directory(test_dir, target_size=(224, 224), batch_size=32, class_mode='binary')

```

```

Epoch 1/10
163/163 ━━━━━━━━━━━ 117s 702ms/step - accuracy: 0.8867 - loss: 0.3717 - val_accuracy: 0.8750 - val_loss: 0.3105
Epoch 2/10
163/163 ━━━━━━━━━━━ 113s 695ms/step - accuracy: 0.9302 - loss: 0.1719 - val_accuracy: 0.6875 - val_loss: 0.8621
Epoch 3/10

```

```
163/163 ————— 113s 690ms/step - accuracy: 0.9360 - loss: 0.1679 - val_accuracy: 0.7500 - val_loss: 0.4512
Epoch 4/10
163/163 ————— 113s 694ms/step - accuracy: 0.9427 - loss: 0.1453 - val_accuracy: 0.8750 - val_loss: 0.3023
Epoch 5/10
163/163 ————— 113s 693ms/step - accuracy: 0.9496 - loss: 0.1232 - val_accuracy: 0.6875 - val_loss: 1.2747
Epoch 6/10
163/163 ————— 112s 690ms/step - accuracy: 0.9549 - loss: 0.1227 - val_accuracy: 0.6875 - val_loss: 0.7850
Epoch 7/10
163/163 ————— 112s 690ms/step - accuracy: 0.9482 - loss: 0.1325 - val_accuracy: 0.8750 - val_loss: 0.2964
Epoch 8/10
163/163 ————— 111s 679ms/step - accuracy: 0.9561 - loss: 0.1109 - val_accuracy: 0.7500 - val_loss: 0.4856
Epoch 9/10
163/163 ————— 111s 682ms/step - accuracy: 0.9542 - loss: 0.1133 - val_accuracy: 0.8125 - val_loss: 0.2943
Epoch 10/10
163/163 ————— 115s 701ms/step - accuracy: 0.9590 - loss: 0.1064 - val_accuracy: 0.8750 - val_loss: 0.2446
Found 624 images belonging to 2 classes.
```

```
test_loss, test_accuracy = model.evaluate(test_generator)
print(f"Test Loss: {test_loss:.4f}, Test Accuracy: {test_accuracy:.4f}")
```